

ResQ-Guard

City-Wide AI Engine for Multi-Camera ANPR, Trajectory Tracking & Urban Traffic Analytics

Smart India Hackathon 2026 • Team Codester • Problem Statement SIH26127

Project Overview

ResQ-Guard, developed under the Smart India Hackathon (SIH) 2026 by Team Codester for problem statement SIH26127, is a centralized AI software platform that transforms a city's existing network of CCTV and ANPR cameras into a single, connected vehicle-intelligence grid. Instead of processing each camera feed in isolation, ResQ-Guard detects and reads every vehicle across the entire network, reconstructs the complete city-wide travel trajectory of any licence plate on a live GIS map, and computes macro traffic-flow analytics and real-time security alerts for law-enforcement and traffic-management authorities. This document explains each feature of the prototype in depth — its functionality, the underlying technology stack, and its operational workflow — with prototype screenshots included for clarity.

Workflow Overview

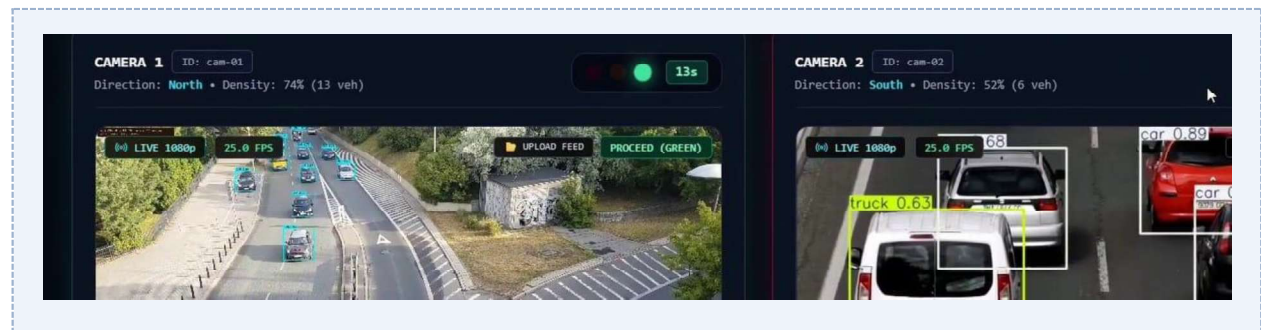
The platform operates as a continuous processing cycle: (1) multi-camera CCTV/ANPR feeds are ingested; (2) vehicles and plates are detected and read; (3) detections are fused across cameras and stored in a spatio-temporal database; (4) trajectories, traffic density, and anomalies are analysed; and (5) the command centre responds with live maps, alerts, and emergency routing. Each stage is described feature-by-feature below.

Features

1. Multi-Camera ANPR & OCR Engine

Functionality: Each camera feed is processed by a YOLOv8 detector that locates vehicles and licence plates, while ByteTrack maintains a stable identity for every vehicle across frames. A dual-engine OCR pipeline (PaddleOCR + EasyOCR) then reads the plate text, cross-validating the two engines to achieve a target accuracy above 90% across challenging real-world conditions such as varying lighting, poor weather, angled shots, motion blur, and dirty or damaged plates.

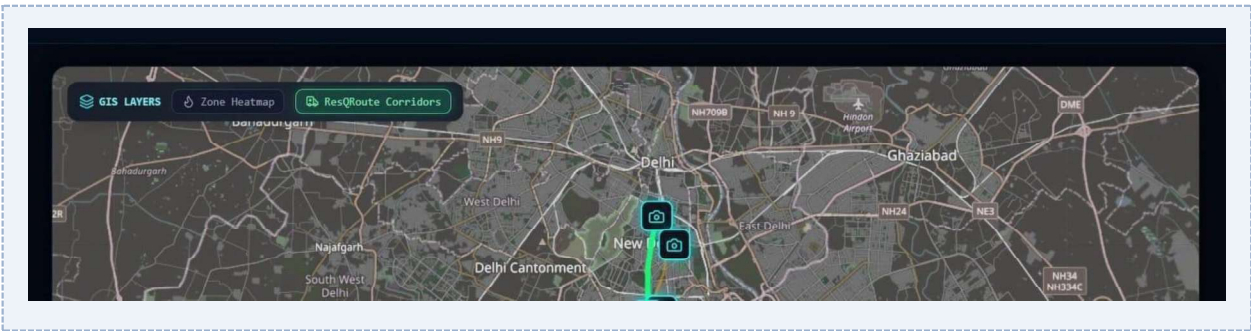
Tech Stack: YOLOv8 • ByteTrack • PaddleOCR • EasyOCR • OpenCV • Python



2. Cross-Camera Trajectory Reconstruction

Functionality: The core innovation of ResQ-Guard: detections of the same plate from geographically distributed, non-overlapping cameras are stitched into one continuous journey. Using spatio-temporal matching, the system rebuilds a vehicle's full movement history — route, direction, and timestamps — and plots it chronologically on a GIS map, letting authorities trace any high-interest vehicle across the entire city.

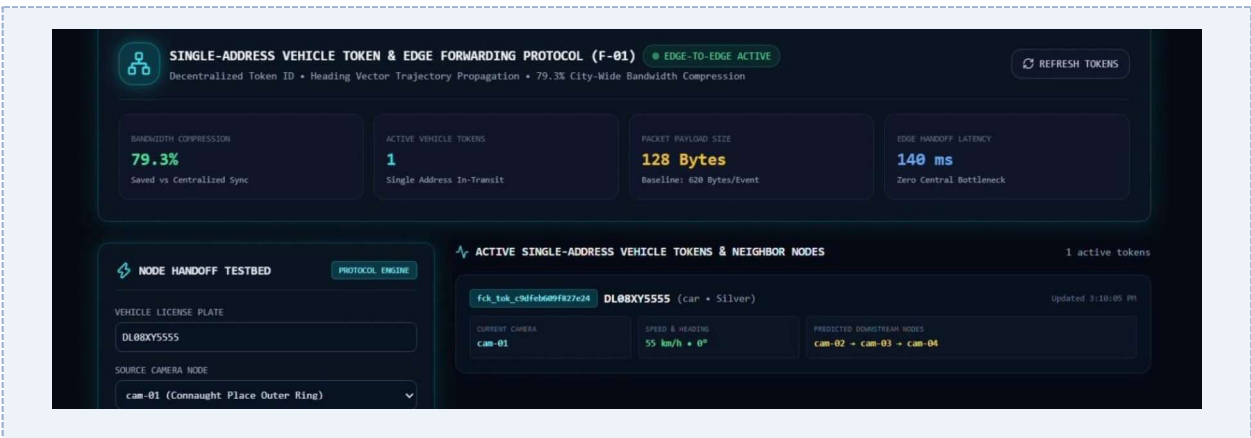
Tech Stack: Python · PostgreSQL + PostGIS · Redis · NetworkX



3. Live GIS Command Map

Functionality: A real-time command-centre dashboard visualises live vehicle movement and the health of every camera on an interactive map. Operators can replay a vehicle's trajectory, open live stream popups for any node, and monitor the whole network at a glance.

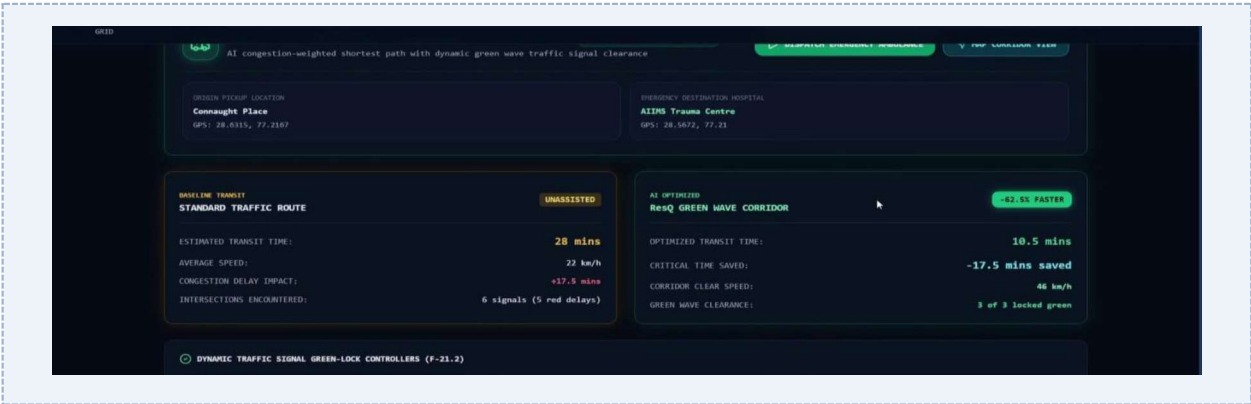
Tech Stack: React · TypeScript · MapLibre GL JS · WebSocket · FastAPI



4. ResQRoute — Emergency Green Corridor

Functionality: A four-stage automated dispatch workflow that computes the fastest route for an emergency vehicle (e.g. an ambulance) and coordinates traffic-signal preemption along that route to clear a continuous 'green corridor', reducing response time in critical situations.

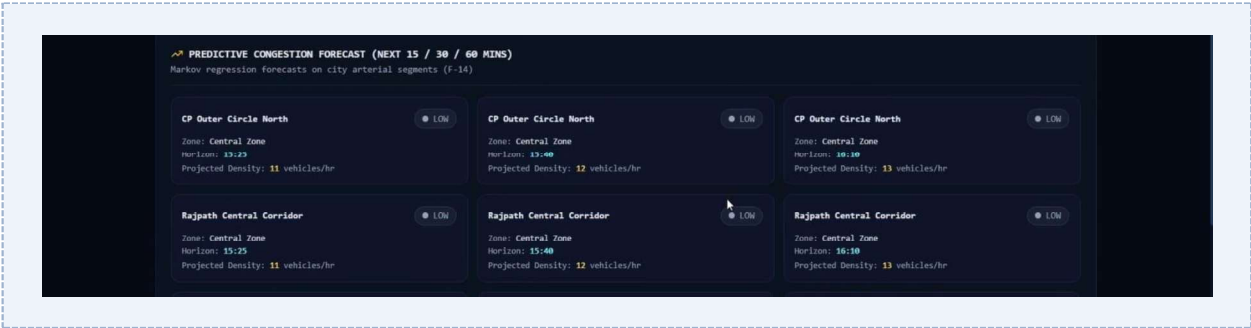
Tech Stack: OSRM · OSMnx / NetworkX · FastAPI · PostGIS



5. Traffic & Security Analytics

Functionality: Aggregated camera data is analysed to compute city-wide traffic dynamics — vehicle density, origin–destination patterns, and congestion bottlenecks — and rendered as live heatmaps. In parallel, the system flags blacklisted vehicles and suspicious route anomalies in real time for security teams.

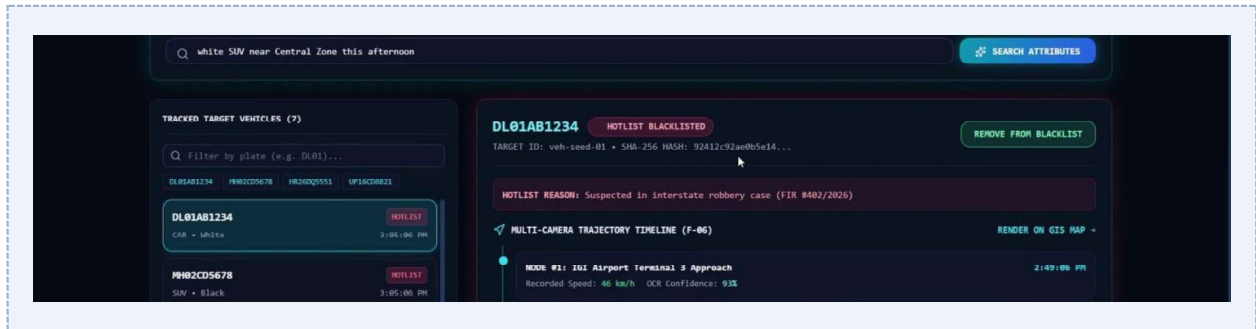
Tech Stack: Python · PostGIS · Redis · MapLibre heatmaps



6. Vehicle Intelligence Dossiers

Functionality: Every tracked vehicle has a per-vehicle case file that consolidates its complete sighting history, reconstructed trajectory, and associated metadata, with the ability for officers to log notes — turning raw detections into an investigation-ready record.

Tech Stack: FastAPI · PostgreSQL · React



7. AI City Assistant

Functionality: A natural-language interface lets operators query traffic and camera data conversationally (for example, 'show all red trucks seen near Sector 12 after 8 pm'). The assistant returns the matching results and can generate exportable reports, making the platform accessible without technical query skills.

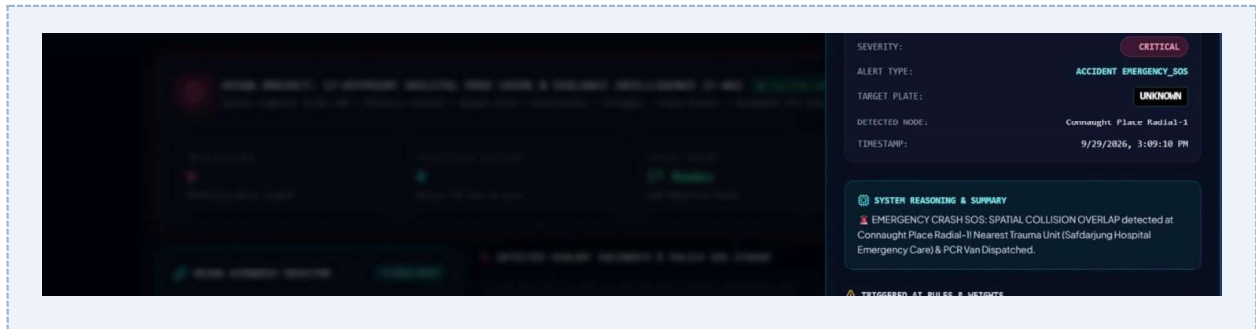
Tech Stack: Python · Natural-language query layer · FastAPI



8. Mobile Patrol App (PWA)

Functionality: A progressive web app equips field officers to scan licence plates on the move, receive dispatch actions, and act on alerts directly from a phone — extending the command centre into the field without a native install.

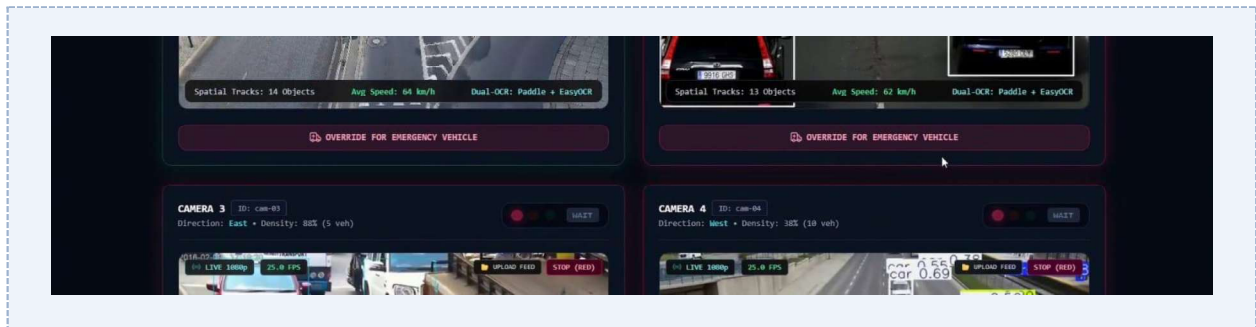
Tech Stack: React PWA · MapLibre GL JS · FastAPI



9. Camera Health Monitoring

Functionality: A live monitoring layer tracks per-camera FPS, latency, and connectivity, with automatic failover detection, so operators immediately know when a node degrades or drops — keeping the intelligence grid reliable.

Tech Stack: FastAPI · Redis · React



Consolidated Technology Stack

Layer	Technologies
Frontend	React · Vite · TypeScript · TailwindCSS · MapLibre GL JS
Backend	FastAPI (async, Python) · WebSocket
Database & Cache	PostgreSQL + PostGIS · Redis
Computer Vision / AI	YOLOv8 · ByteTrack · PaddleOCR · EasyOCR · OpenCV
GIS & Routing	OSRM · OSMnx / NetworkX
Deployment	Docker · docker-compose

Deployment & Resources

The entire platform is containerised for reproducible deployment using Docker and docker-compose, allowing the vision pipeline, backend, database, and frontend to be brought up together on any host.

GitHub Repository: github.com/itssouradip34/ResQ-Guard

Demo Video:

Live Website: [ResQ-Guard | City-Wide AI ANPR & Vehicle Intelligence Platform](https://resq-guard.com)